

DataSet2 — Survival Analysis

Survivorship of a census-scored lifespan cohort. Curves are Kaplan-Meier estimates over individuals; individuals unaccounted for at the final census are right-censored unless that policy is switched off in the experiment's configuration.

Experiment type:	Standard Lifespan
Data file:	DataSet2.xlsx
Generated:	2026-08-27 21:38
Factors:	Sex, Density
Treatments:	4
Individuals:	5600
Deaths:	5402
Censored:	198 (3.5%)
Exclusion group:	none
Censoring policy:	unaccounted individuals censored

Omnibus log-rank: treatments differ ($p = <0.0001$).

Experiment summary

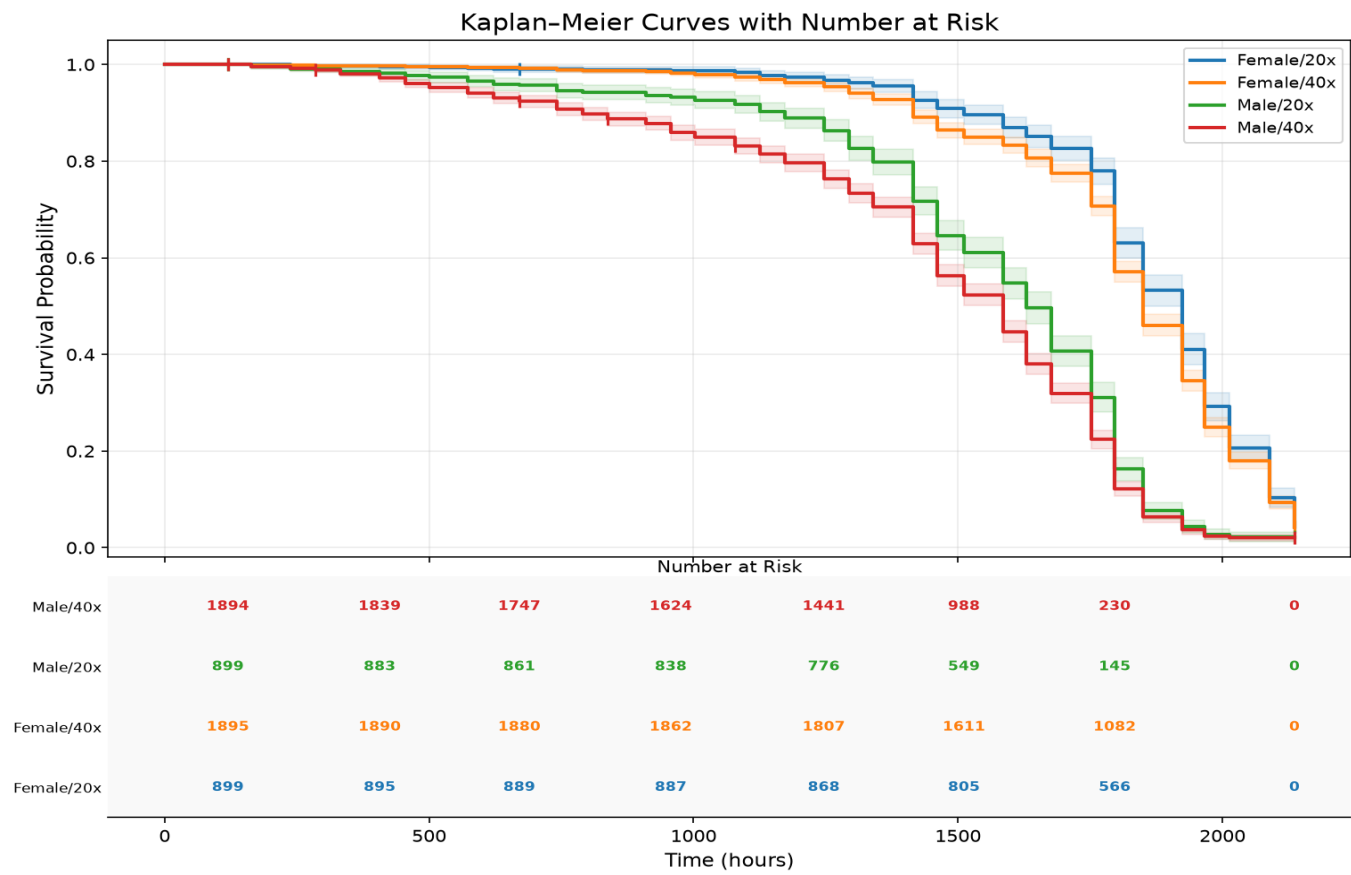
Per-treatment summary

treatment	n_individuals	n_deaths	n_censored	pct_censored
Female/20x	900	859	41	4.600
Female/40x	1900	1815	85	4.500
Male/20x	900	878	22	2.400
Male/40x	1900	1850	50	2.600

Observation window 119.62–2136.14, 200 chamber(s).

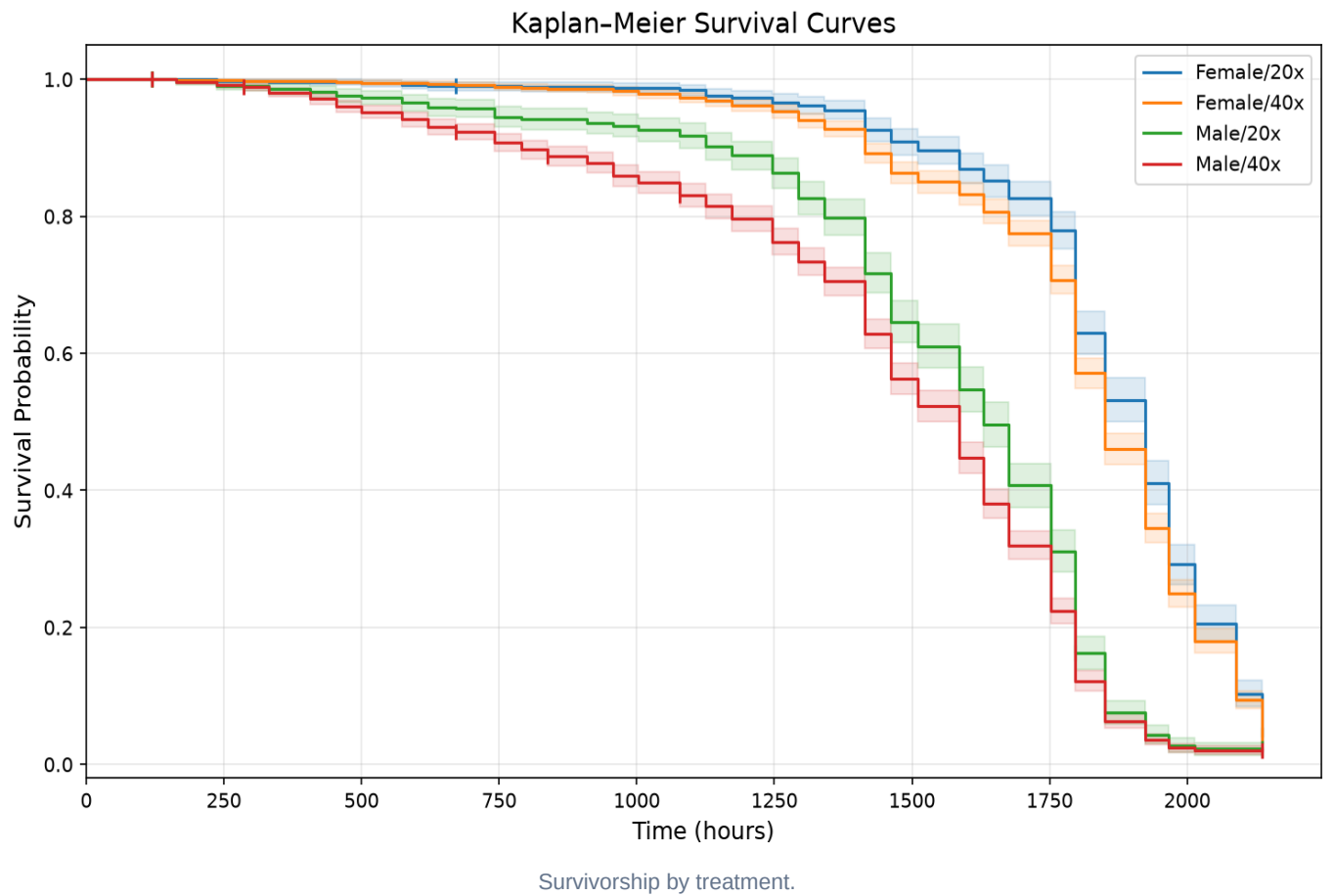
Survivorship figures

KM curves with at-risk table — headline figure

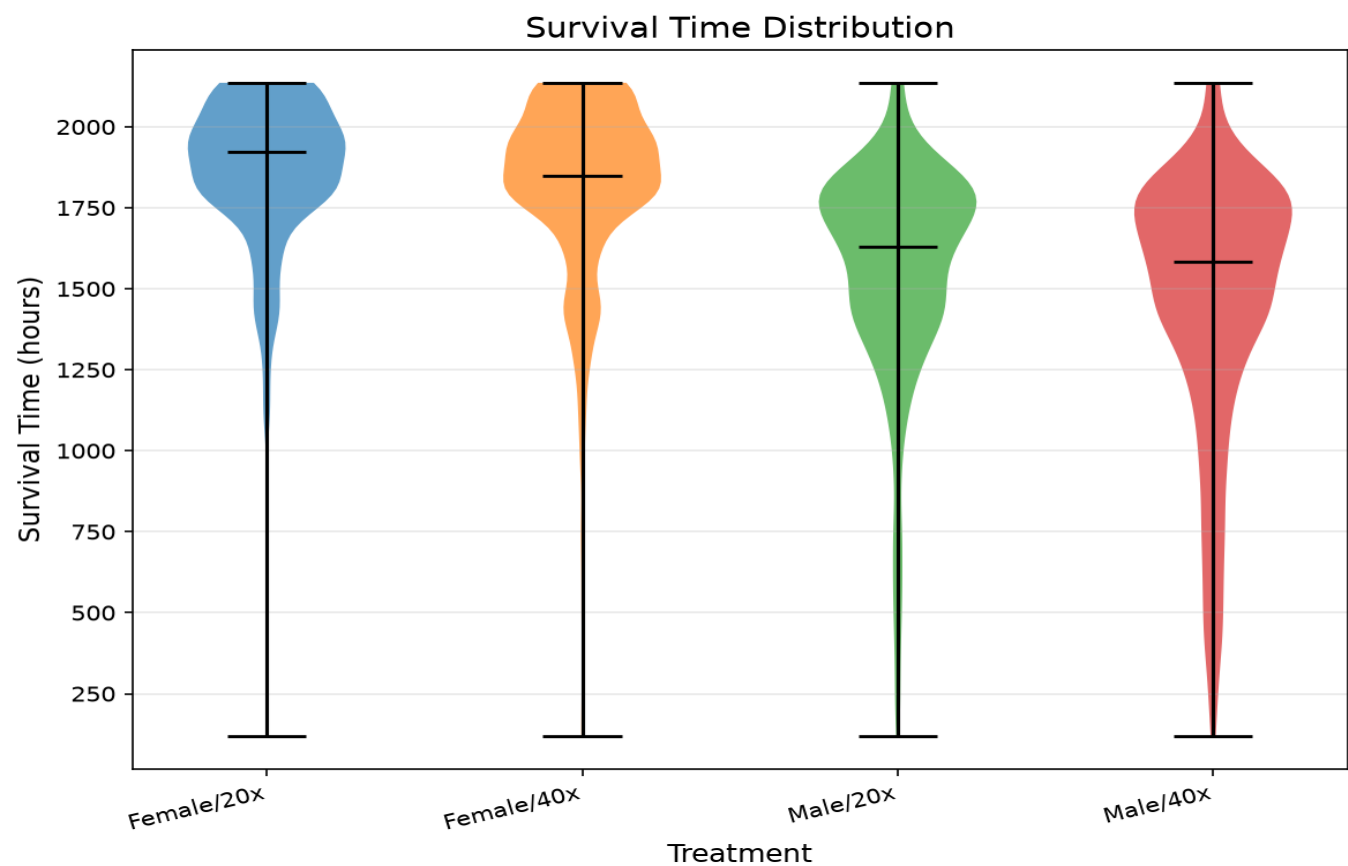


Survivorship with the number at risk beneath the axis.

Kaplan-Meier curves

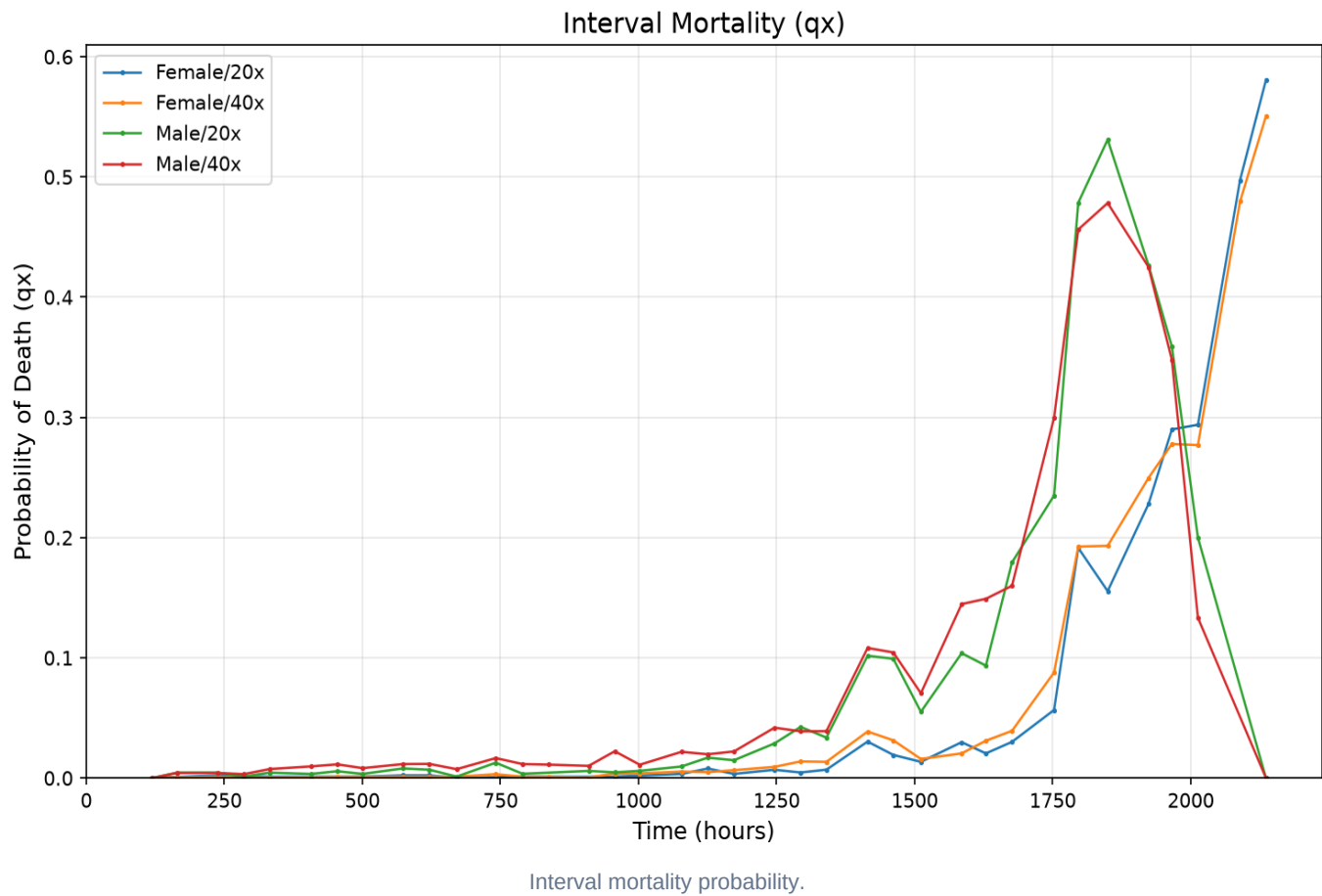


Lifespan distribution

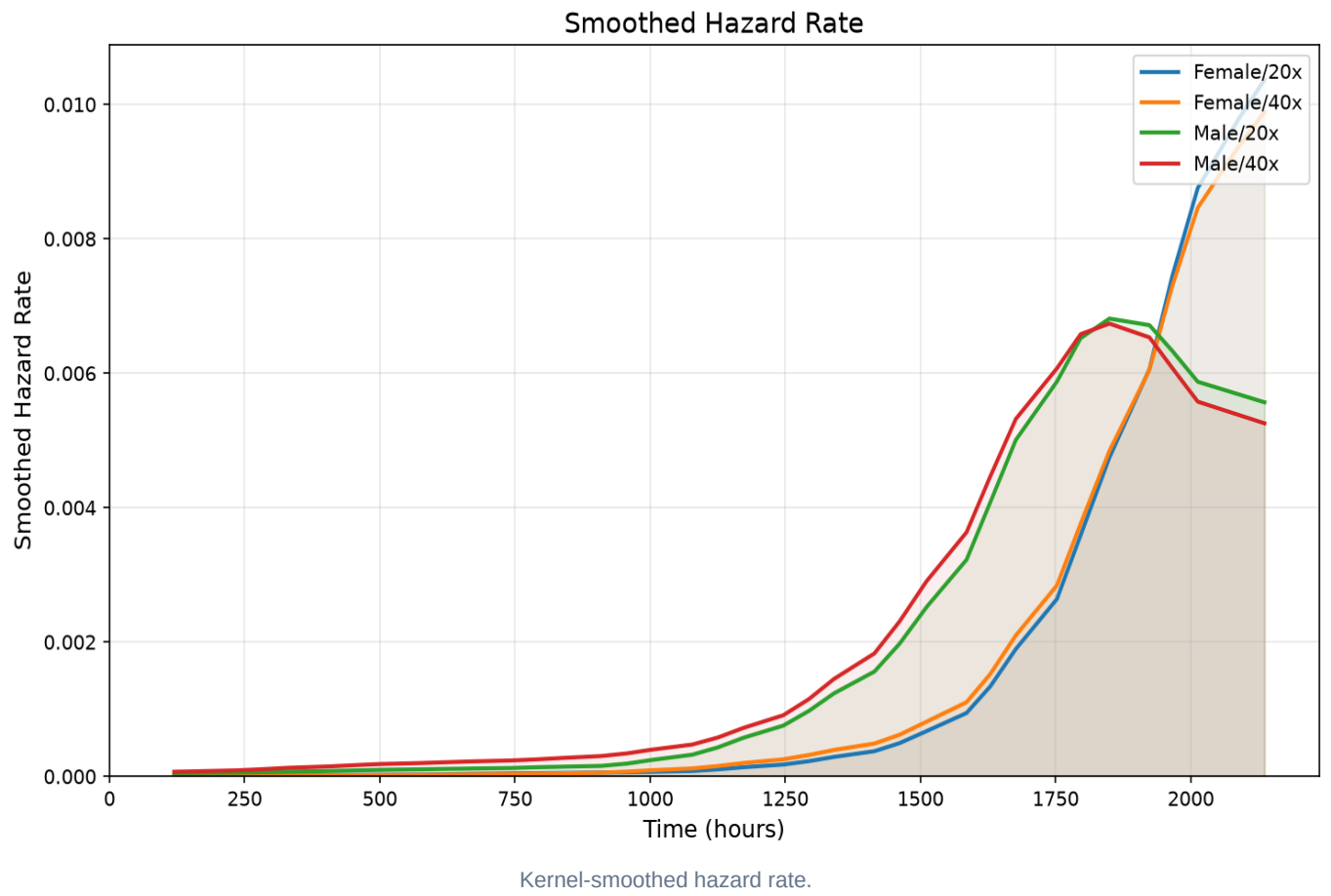


Distribution of individual lifespans by treatment.

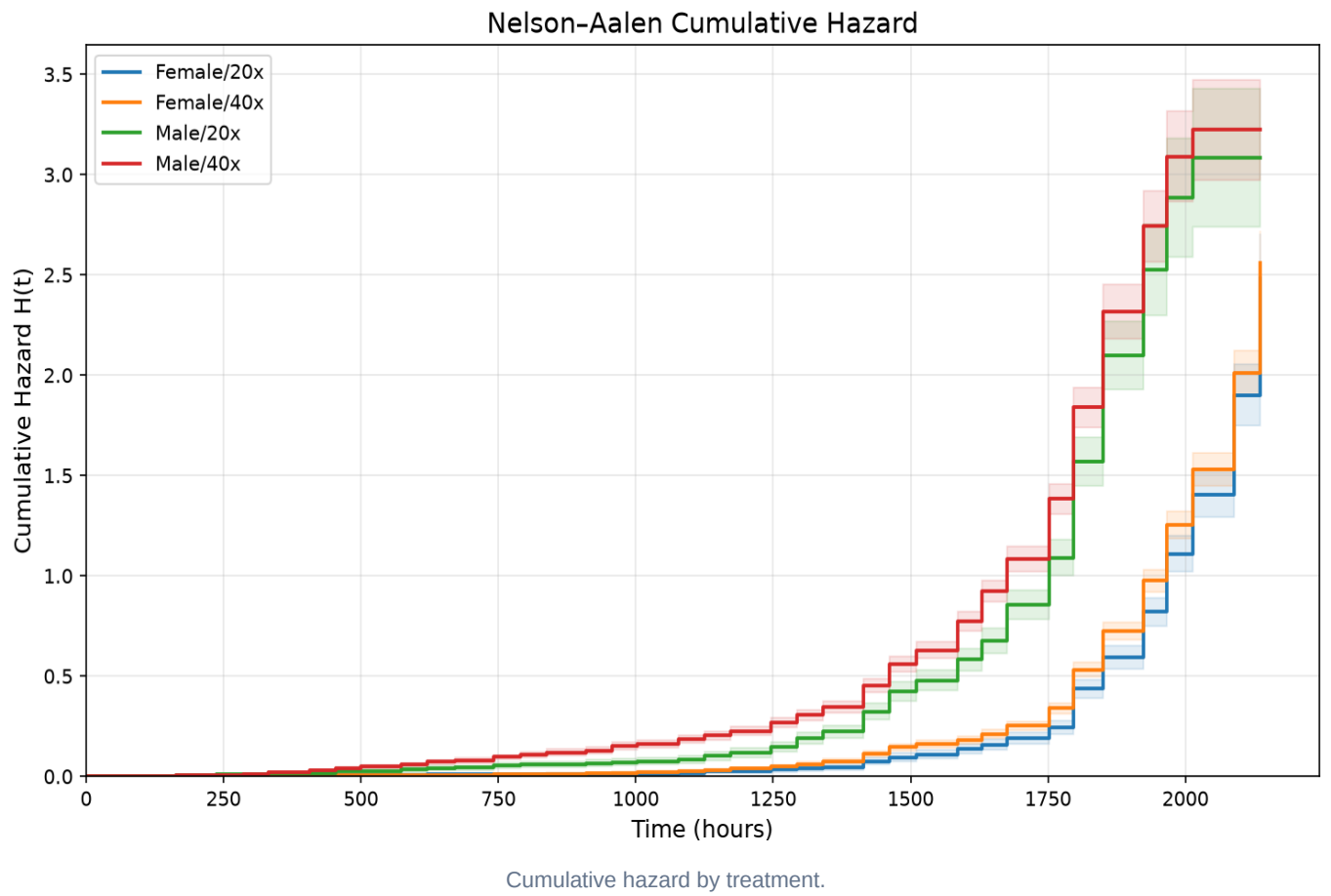
Mortality (qx)



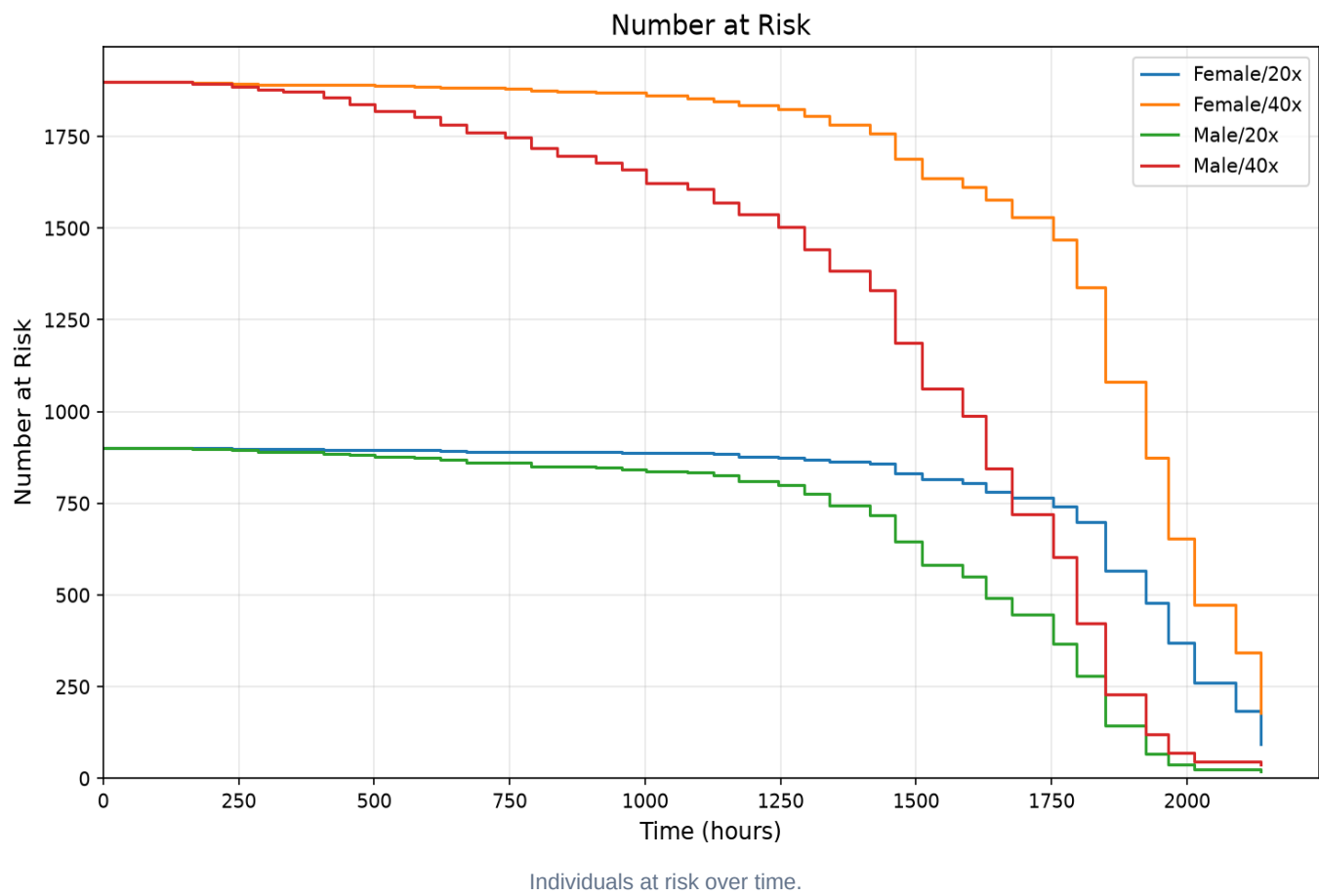
Smoothed hazard



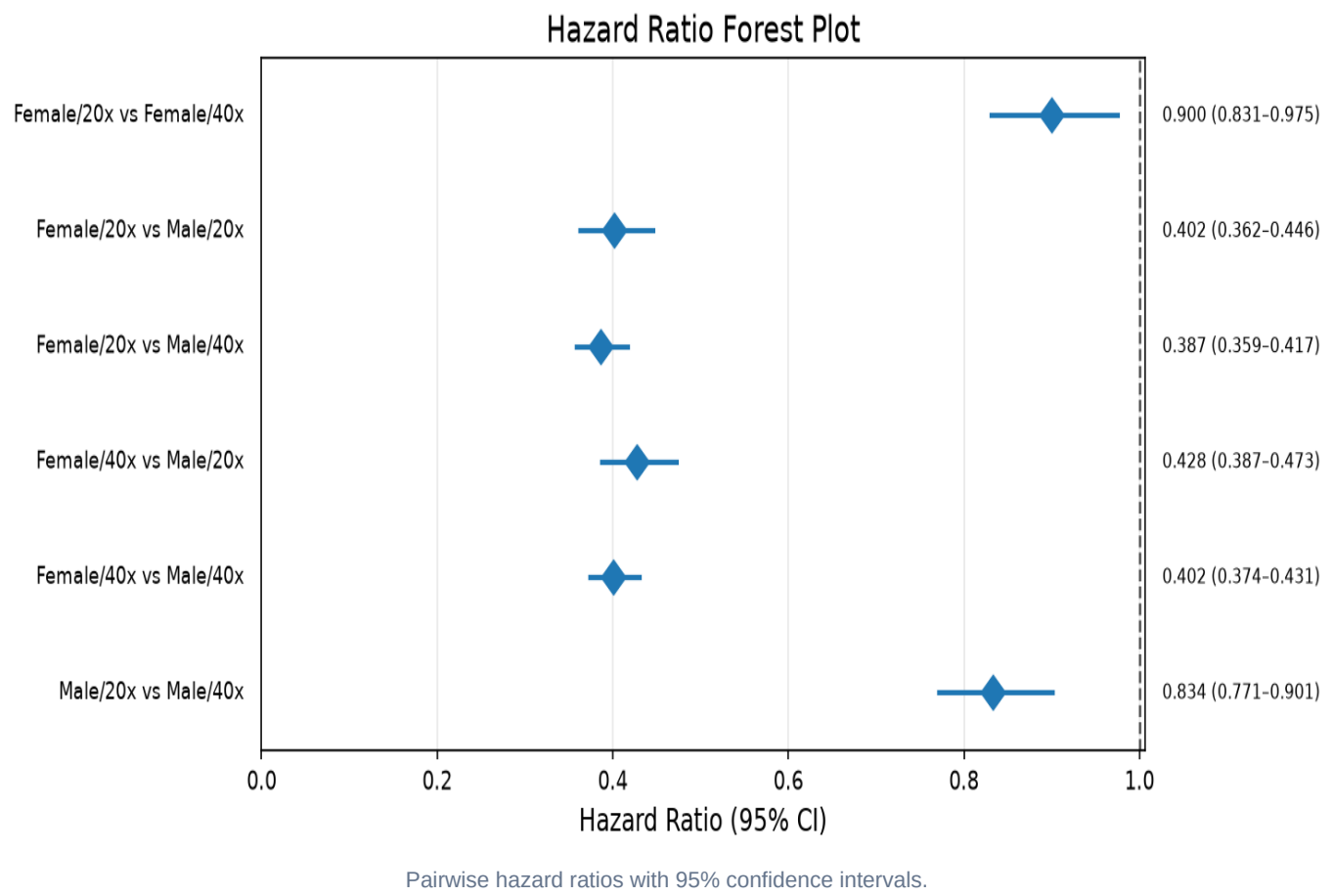
Nelson-Aalen cumulative hazard



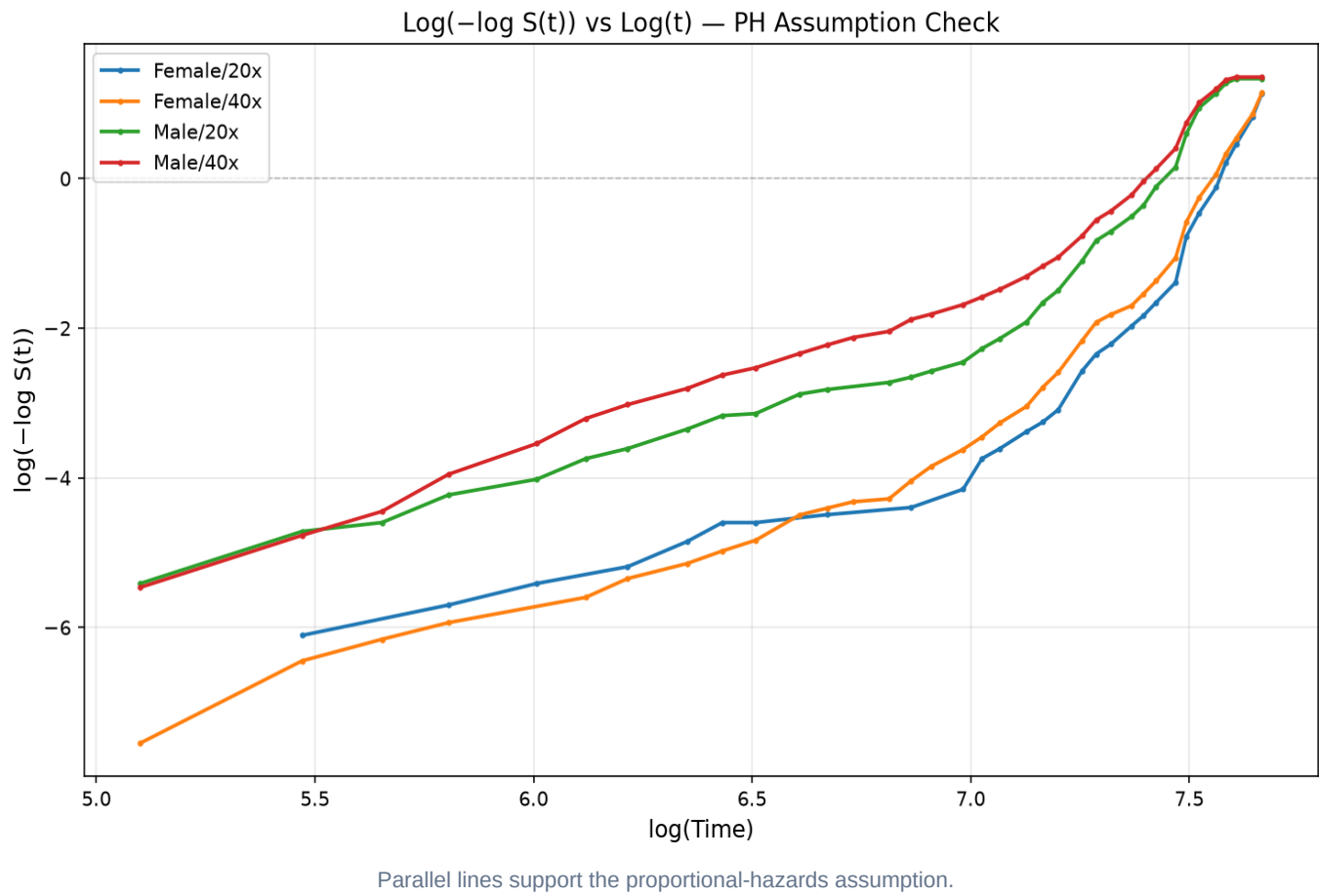
Number at risk



Hazard-ratio forest



Log-log diagnostic



Lifespan statistics

Median survival

treatment	median_survival
Female/20x	1923.479
Female/40x	1849.431
Male/20x	1628.977
Male/40x	1585.109

Mean survival

treatment	rmst	restriction_time
Female/20x	1825.136	2136.138
Female/40x	1785.516	2136.138
Male/20x	1529.745	2136.138
Male/40x	1431.423	2136.138

Lifespan statistics by treatment

group	n	n_deaths	n_censored	mean_rmst	median	t_max
Female/20x	900	859	41	1825.140	1923.480	2136.140
Female/40x	1900	1815	85	1785.520	1849.430	2136.140
Male/20x	900	878	22	1529.740	1628.980	2136.140
Male/40x	1900	1850	50	1431.420	1585.110	2136.140

Lifespan statistics by factor level

group	n	n_deaths	n_censored	mean_rmst	median	t_max
Sex=Female	2800	2674	126	1798.370	1849.430	2136.140
Sex=Male	2800	2728	72	1463.150	1585.110	2136.140
Density=20x	1800	1737	63	1677.540	1796.280	2136.140
Density=40x	3800	3665	135	1608.720	1752.370	2136.140

Survival quantiles

treatment	S=90%	S=75%	S=50%	S=25%	S=10%
Female/20x	1511.224	1796.282	1923.479	2012.853	2136.138
Female/40x	1414.742	1752.370	1849.431	1965.732	2089.093
Male/20x	1172.736	1414.742	1628.977	1796.282	1849.431
Male/40x	789.977	1293.467	1585.109	1752.370	1849.431

Survival comparisons

Overall comparison

Test	chi²	df	p	
Omnibus log-rank	1554.87	3	<0.0001	***

Pairwise log-rank

group1	group2	observed_1	expected_1	chi2	p_value	df	p_bonferroni	significant_0.05
Female/20x	Female/40x	859.000	921.307	8.241	0.004	1	0.025	True
Female/20x	Male/20x	859.000	1231.094	498.050	0.000	1	0.000	True
Female/20x	Male/40x	859.000	1477.824	752.562	0.000	1	0.000	True
Female/40x	Male/20x	1815.000	2230.979	554.198	0.000	1	0.000	True
Female/40x	Male/40x	1815.000	2600.681	1025.915	0.000	1	0.000	True
Male/20x	Male/40x	878.000	989.550	24.441	0.000	1	0.000	True

Pairwise Gehan-Wilcoxon

group1	group2	chi2	p_value	df	p_bonferroni	significant_0.05
Female/20x	Female/40x	14.331	0.000	1	0.001	True
Female/20x	Male/20x	523.649	0.000	1	0.000	True
Female/20x	Male/40x	764.381	0.000	1	0.000	True
Female/40x	Male/20x	585.387	0.000	1	0.000	True
Female/40x	Male/40x	1082.786	0.000	1	0.000	True
Male/20x	Male/40x	36.552	0.000	1	0.000	True

Pairwise hazard ratios

group1	group2	hazard_ratio	hr_ci_lo	hr_ci_hi
Female/20x	Female/40x	0.900	0.831	0.975
Female/20x	Male/20x	0.402	0.362	0.446
Female/20x	Male/40x	0.387	0.359	0.417
Female/40x	Male/20x	0.428	0.387	0.473
Female/40x	Male/40x	0.402	0.374	0.431
Male/20x	Male/40x	0.834	0.771	0.901

Data quality

No chambers were excluded from this analysis.

Parametric model fits

Parametric model	AIC
results_by_treatment	—
aic_comparison	treatment model aic log_likelihood median_survival 0 Female/20x Weibull 11940.51 -5968.2568 1880.65 1 Female/20x Log-Normal 12692.05 -6344.0231 1834.28 2 Female/20x Log-Logistic 12174.54 -6085.2693 1880.53 3 Female/40x Weibull 25582.86 -12789.4291 1843.01 4 Female/40x Log-Normal 26881.66 -13438.8292 1791.09 5 Female/40x Log-Logistic 26069.57 -13032.7858 1841.58 6 Male/20x Weibull 12799.89 -6397.9467 1576.51 7 Male/20x Log-Normal 13499.81 -6747.9053 1499.72 8 Male/20x Log-Logistic 13079.85 -6537.9233 1583.95 9 Male/40x Weibull 27612.22 -13804.1099 1473.14 10 Male/40x Log-Normal 28739.29 -14367.6425 1378.18 11 Male/40x Log-Logistic 28244.27 -14120.1364 1479.89
best_model_per_treatment	—